# Questions

Q1.

A scientist records the sounds made by a dolphin as it swims towards him.

The dolphin stops swimming but continues to emit the sounds.

Describe what happens to the frequency and wavelength of the sounds when the dolphin stops swimming.

**(2)**

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**(Total for question = 2 marks)**

Q2.

There have been a number of theories about the origin and development of the Universe.

One theory about the origin and development of the Universe is The Big Bang Theory.

Discuss **two** pieces of evidence that support The Big Bang Theory.

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**(Total for question = 6 marks)**

Q3.

(i) A star with a mass very much larger than the Sun

**(1)**

☒ **A** has a longer main sequence than the Sun and ends as a white dwarf

☒ **B** has a longer main sequence than the Sun and ends as a black hole

☒ **C** has a shorter main sequence than the Sun and ends as a white dwarf

☒ **D** has a shorter main sequence than the Sun and ends as a black hole

(ii) Which row has two correct statements about black holes?

**(1)**

| | the gravitational field of a black hole | a black hole is formed when |
|---|---|---|
| ☐ A | allows only electromagnetic radiation to escape | a nebula collapses |
| ☐ B | allows nothing to escape | a very large star collapses |
| ☐ C | allows nothing to escape | a nebula collapses |
| ☐ D | allows only electromagnetic radiation to escape | a very large star collapses |

**(Total for question = 2 marks)**

Q4.

Figure 4 is a diagram giving some information about main sequence stars.
Luminosity is a measure of how bright something is.
An increase in luminosity means an increase in brightness.

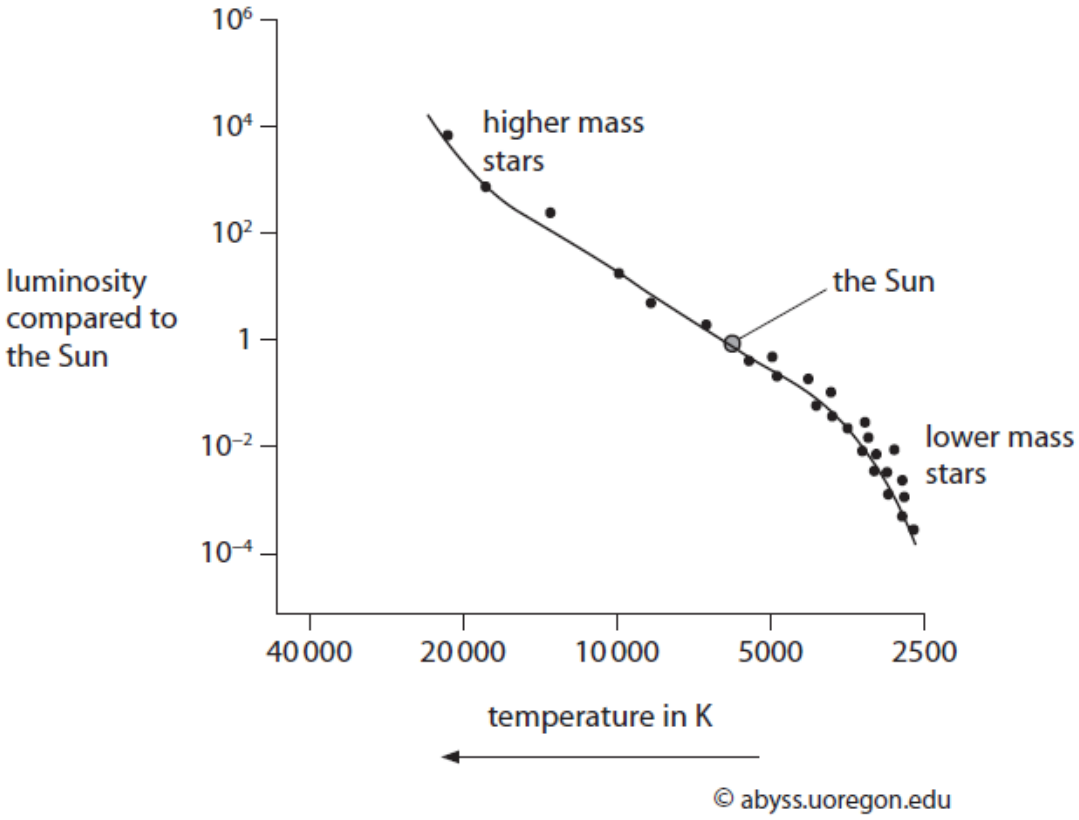


**Figure 4**

(i) Estimate the temperature of the Sun.

**(1)**

temperature of the Sun = ..... K

(ii) State how the brightness of a main sequence star changes with its temperature.

**(1)**

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(iii) State how the brightness of a main sequence star changes with its mass.

**(1)**

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**(Total for question = 3 marks)**

Q5.

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

Observations of the expanding Universe have shown that the further away a galaxy is from the Earth, the faster the galaxy is moving away from the Earth.

Figure 4 shows how the velocity of galaxies is related to their distance from the Earth.

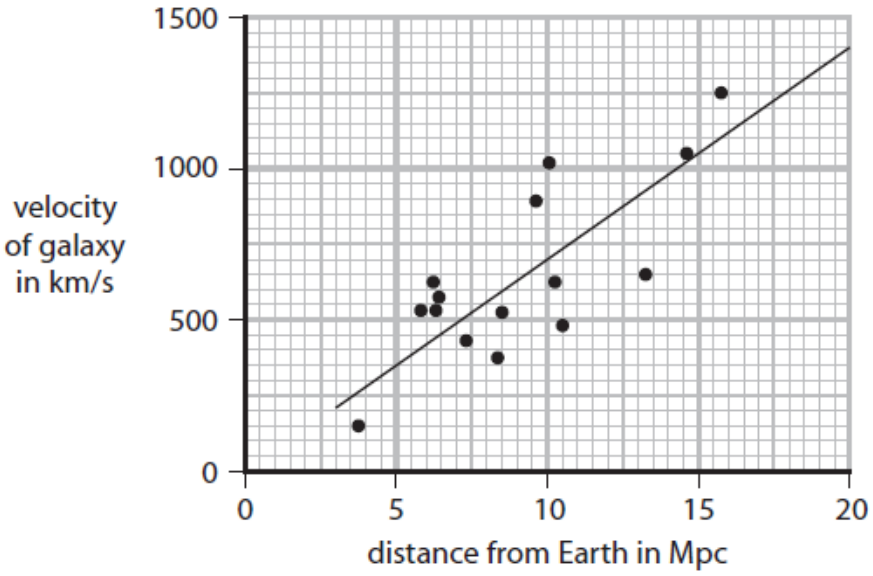


**Figure 4**

Mpc is a unit of distance used for large distances in space.

(i) Use Figure 4 to estimate the velocity of a galaxy that is 15 Mpc away from the Earth.

**(1)**

velocity = ..... km/s

(ii) Calculate the gradient of the line shown in Figure 4.

State the unit.

**(3)**

gradient = ..... unit .....

(iii) The gradient of the line in Figure 4 can be used to estimate the age of the Universe.

Explain why the gradient of the line in Figure 4 can only provide an **estimate** of the age of the Universe.

**(2)**

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**(Total for question = 6 marks)**

Q6.

**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

Which of these provides evidence that the Universe had a beginning?

**(1)**

☐ **A** the discovery of other galaxies

☐ **B** the discovery of the moons of Jupiter

☐ **C** the discovery of planets orbiting distant stars

☐ **D** the discovery of cosmic microwave background (CMB) radiation

**(Total for question = 1 mark)**

Q7.

Figure 1 shows the spectrum of an element detected in the light from a distant galaxy, from a nearby galaxy and from a source on Earth.

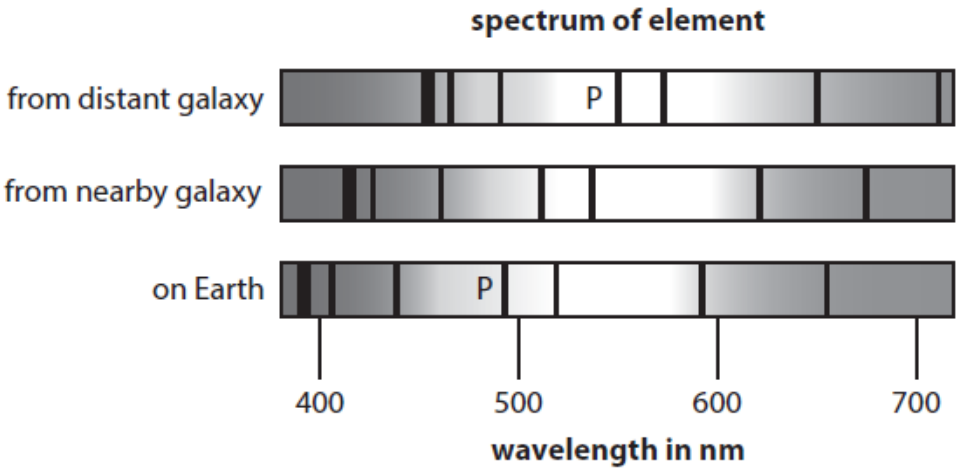


**Figure 1**

(i) Estimate the difference between the wavelength of line P in the spectrum from the distant galaxy and the wavelength of line P in the spectrum on Earth.

**(1)**

difference in wavelength = ..... nm

(ii) Scientists have discovered that light from almost all distant galaxies has spectral lines shifted towards the red end of the spectrum.

Explain how red shift in light, received from galaxies at different distances from the Earth, supports the idea that the Universe is expanding.

**(3)**

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**(Total for question = 4 marks)**

Q8.

A nebula may evolve into a main sequence star, such as the Sun.

Explain how a nebula may evolve into a main sequence star.

**(3)**

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**(Total for question = 3 marks)**

Q9.

The Big Bang theory gives an explanation for the origin of the Universe.

Explain how evidence supports the ideas that

- the Universe is expanding
- the Universe began at a single point.

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**(Total for question = 6 marks)**

Q10.

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

Evidence that the Universe is still expanding comes from observations of light from distant galaxies.

Describe how light from distant galaxies shows that the Universe is still expanding.

**(2)**

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**(Total for question = 2 marks)**

Q11.

The wavelength of light emitted from distant galaxies is different when the light is detected on Earth.

Explain how this difference in wavelength shows that the Universe is expanding.

**(2)**

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**(Total for question = 2 marks)**

## Mark Scheme

Q1.

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | an answer that provides a description by making reference to:<br>• the frequency is lower (1)<br>• the wavelength is longer (1) | award one mark for doppler shift if no other mark is awarded | **(2)** |

Q2.

| Question number | Indicative content | Mark |
|---|---|---|
| * | Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme.<br>The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant.<br>AO1<br>Red-shift Description<br>• increase in wavelength (detected) / decrease in frequency<br>• Doppler effect (for light)<br>• light from distant galaxies/stars<br><br>Red-shift Support of Big Bang Theory<br>• increase corresponds to moving away from each other<br>• greater red-shift means moving (away) faster<br>• more distant galaxies show greater red-shift<br>• Universe is expanding / galaxies moving away<br><br>CMB(R) Description<br>• microwave radiation<br>• detected everywhere<br><br>CMB(R) Support of Big Bang Theory<br>• left over from the start of the Universe / gamma rays released at Big Bang<br>• indicates cooling<br>• Universe had a (definite) beginning | **(6) AO1** |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | • No rewardable material. relevant |
| Level 1 | 1-2 | • Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1)<br>• Presents an explanation with some structure and coherence. (AO1) |
| Level 2 | 3-4 | • Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1)<br>• Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1) |
| Level 3 | 5-6 | • Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1)<br>• Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1) |

| Level | Mark | Additional Guidance | General additional guidance – the decision within levels<br><br>e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level. |
|---|---|---|---|
| | 0 | No rewardable material. | |
| Level 1 | 1–2 | Additional guidance<br><br>isolated statements:<br>identifies 2 of:<br>aspects of BB theory OR<br>red shift OR<br>CMBR | Possible candidate responses<br><br>BB says Universe is expanding AND<br>CMBR is evidence<br>**OR**<br>Red Shift and CMBR are evidence |
| Level 2 | 3–4 | Additional guidance<br><br>limited explanation:<br><br>**EITHER**<br>identifies & describes 1 piece of evidence;<br>**OR**<br>identifies & links 1 piece of evidence to the aspect of the BB theory it supports<br><br>**AND**<br><br>identifies the other piece of evidence | Possible candidate responses<br><br>Red Shift (is the) increase in wavelength (detected) in light from distant galaxies.<br>CMBR (is evidence)<br><br>OR<br><br>Red Shift indicates (all) galaxies are moving away from each other.<br>CMBR (is evidence) |
| Level 3 | 5–6 | Additional guidance<br><br>detailed explanation:<br><br>identifies **AND** describes 1 piece of evidence **AND** links it to the aspect of the BB theory it supports.<br><br>**AND**<br><br>identifies the other piece of evidence | Possible candidate responses<br><br>Red Shift (is the) increase in wavelength (detected) in light from distant galaxies which indicates (all) galaxies are moving away from each other.<br>CMBR (is evidence)<br><br>OR<br><br>CMBR microwave radiation detected everywhere which indicates the Universe had a beginning.<br>Redshift (is evidence) |

Q3.

| Question number | Answer | Mark |
|---|---|---|
| (i) | B | (1) |

| Question number | Answer | Mark |
|---|---|---|
| (ii) | B | (1) |

Q4.

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | accept any temperature between 5500 and 7500 (K) (1) | | (1) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | the greater the temperature the higher the brightness | or reverse argument<br><br>allow luminosity for brightness<br>allow heat for temperature | (1) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (iii) | the greater the mass the higher the brightness | or reverse argument<br>allow luminosity for brightness<br>allow bigger/ heavier for greater mass in this context | (1) |

Q5.

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | 1050 ± 20 (km/s) | | (1)<br>AO3 |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | attempt at $\frac{\Delta y}{\Delta x}$ (1)<br><br>evaluation (1)<br><br>70 ± 5<br><br>unit (1)<br><br>km/s/Mpc | could be seen on graph<br><br>award 2 marks for correct answer without working<br><br>independent mark<br><br>km/s Mpc<br>$s^{-1}$ or per second | (3)<br>AO3 |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (iii) | an explanation linking:<br><br>points are scattered widely about the line (on graph) (1)<br><br>giving wide range of possible gradients (1) | there are many possible best fit lines | (2)<br>AO3 |

Q6.

| Question number | Answer | Mark |
|---|---|---|
| | **The only correct answer is**<br>**D** the discovery of cosmic microwave background (CMB) radiation<br><br>**A** is not correct because it does not indicate the Universe had a beginning<br><br>**B** is not correct, it is evidence against the geocentric model of the Universe<br><br>**C** is not correct, it is evidence for other solar systems | (1)<br>AO1 |

Q7.

| Question | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | 50-70 (nm) inclusive | | 1<br>AO1.2 |

| Question | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | explanation linking<br><br>(red shift means) galaxies must be moving away from Earth /us (1)<br><br>more distant galaxies have bigger red shift (1)<br><br>more distant galaxies moving faster (1) | <br><br>galaxies getting further apart<br><br>ORA | 3<br>AO1.1 |

Q8.

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation to include any three from:<br><br>nebula as a cloud of gas/dust (1)<br><br>gas / atoms pulled together / towards each other (1)<br><br>by gravitational force (1)<br><br>temperature increase (1)<br><br>hot enough for nuclear fusion (1) | <br><br><br>density increase<br><br>gravity (acting)<br><br>(resultant) heating | (3) |

Q9.

| Question Number | Answer | Mark |
|---|---|---|
| | Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme.<br><br>The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. | **(6)** |

| | | |
|---|---|---|
| | **AO1 strand 1 (6 marks)**<br><br>**evidence for expansion**<br><br>• red shift<br>• light from distant galaxies/stars<br>• shifted to red side of em spectrum<br>• (observed) wavelength of light is longer<br>• showing source moving away<br>• (nearly) all galaxies show this<br>• Doppler effect<br>• (expanding at an increasing rate / dark energy)<br><br>**evidence for beginning at a point**<br><br>• CMBR<br>• microwave radiation left over from beginning<br>• the increase of recessional velocities with distance - (extrapolating)<br>• microwaves because of cooling<br>• detected from all over the sky | |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable material. |
| Level 1 | 1-2 | • Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1)<br>• Presents an explanation with some structure and coherence. (AO1) |
| Level 2 | 3-4 | • Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1)<br>• Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1) |
| Level 3 | 5-6 | • Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1)<br>• Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1) |

| Summary for guidance | | | |
|---|---|---|---|
| **Level** | **Mark** | **Additional Guidance** | **General additional guidance – the decision within levels**<br><br>e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level. |
| | 0 | No rewardable material. | |
| Level 1 | 1–2 | Additional guidance<br><br>some element of physics about the expansion / single point origin | Possible candidate responses<br><br>red shift shows expansion<br><br>or<br><br>CMBR connected with beginning |
| Level 2 | 3–4 | Additional guidance<br><br>more detail about one piece of evidence or basic detail about two pieces of evidence | Possible candidate responses<br><br>red shift connected with galaxies moving away (from earth observer)<br><br>CMBR to do with release of radiation at the beginning |
| Level 3 | 5–6 | Additional guidance<br><br>Understanding is detailed and fully developed.<br><br>includes detail about both pieces of evidence<br><br>(one may be stronger than the other but both should feature for level 3) | Possible candidate responses<br><br>red shift connected with galaxies moving away (from earth observer) with the further they are away the faster they are moving away<br><br>CMBR to do with release of (leftover) radiation at the beginning  microwaves because of cooling – detected from all over the sky |

Q10.

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | A description to include:<br><br>wavelength (of the light) (1)<br><br>appears to increase (1) [increase must be linked with wavelength] | <br><br>Red shift/Doppler effect<br><br>(Red shift) shows galaxy moving away<br><br>accept answers in terms of frequency | **(2)**<br>**AO1** |

Q11.

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| | (there is) an increase in wavelength (of light) (1)<br><br>shows galaxies are moving away (from Earth) (1) | allow wavelength stretches<br>allow red shift<br>ignore shift to red end of spectrum<br><br>ignore objects / stars / planets | **(2)**<br>**AO1** |